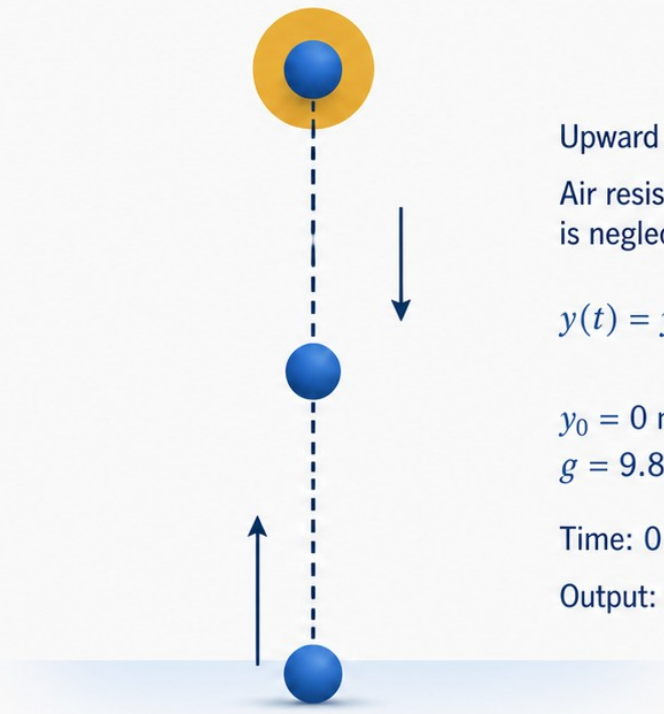


Physics to Arrays and Plots

From a familiar motion model to trustworthy computational evidence.

- A familiar motion model becomes computational evidence.
- Model $\rightarrow$ arrays $\rightarrow$ plot $\rightarrow$ validation $\rightarrow$ interpretation.



Upward is positive.

Air resistance is neglected.

$$y(t) = y_0 + v_0 t - \frac{1}{2} g t^2$$

$$y_0 = 0 \text{ m}, v_0 = 20 \text{ m s}^{-1}, g = 9.81 \text{ m s}^{-2}$$

Time: 0:0.1:4 s

Output: height y versus time t



1. Physical Model

Understand and state the model.



2. Arrays

Define variables and create arrays.



3. Plot

Visualize height versus time.



4. Validation

Check the boundary:
 $y(0) = 0 \text{ m} \Rightarrow y_m(1)$



5. Interpretation

Explain the result in physical terms.

Before MATLAB, sketch what must happen

START
 $y = 0 \text{ m}$



RISING
 y increases



PEAK
one maximum



FALLING
 y decreases



CHECK
compare with
 $y = 0 \text{ m}$



Prediction before computation

- At $t = 0 \text{ s}$, $y = 0 \text{ m}$
- As t increases, y rises to one maximum
- After the peak, y decreases
- Near the end, compare with $y = 0 \text{ m}$

Prediction first

Sketch the shape before coding.

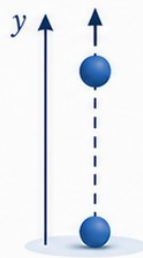
Blue = motion | Green = expected check | Yellow = prompt

Decide what the computer must remember

- Scalars store fixed values such as y_0 , v_0 , and g .
- An array stores many chosen times.
- Every stored quantity needs a meaning and unit.

Physical model (vertical motion)

$$y(t) = y_0 + v_0 t - \frac{1}{2} g t^2$$



$$\begin{aligned} y_0 &= 0 \text{ m} \\ v_0 &= 20 \text{ m s}^{-1} \\ g &= 9.81 \text{ m s}^{-2} \\ t &= 0:0.1:4 \text{ s} \end{aligned}$$

Upward is positive.
Air resistance is neglected.

	Quantity	MATLAB name	Storage form	Meaning	Unit
SCALARS (fixed values)	y_0	y0_m	scalar	initial height (launch height)	m
	v_0	v0_mps	scalar	initial upward speed	m s ⁻¹
	g	g_mps2	scalar	acceleration due to gravity (positive number)	m s ⁻²
ARRAYS (many values)	t	t_s	1 × N array	chosen time samples	s
	$y(t)$	y_m	1 × N array	height at each time in t_s	m



Blue = scalars (fixed values)

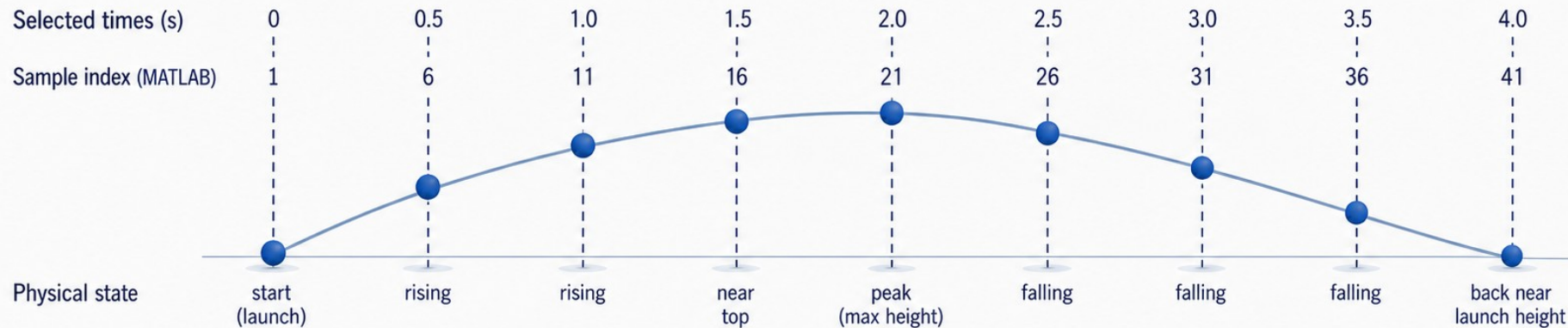


Green = arrays (many values)



Hollow circle = main output we will compute and plot

Sampling is a choice, not the motion itself



Array construction — Method 1

```
t_s = 0:0.1:4;
```

Endpoints: 0 s ————— 4 s

Array construction — Method 2

```
t_s = linspace(0,4,41);
```

Endpoints: 0 s ————— 4 s

41
samples

Key points

- `t_s = 0:0.1:4` and `linspace(0,4,41)` make 41 time samples.
- The endpoints and count match.
- Sampling changes what we see, not the underlying equation.



Model (continuous idea): $y(t) = y_0 + v_0 t - \frac{1}{2} g t^2$

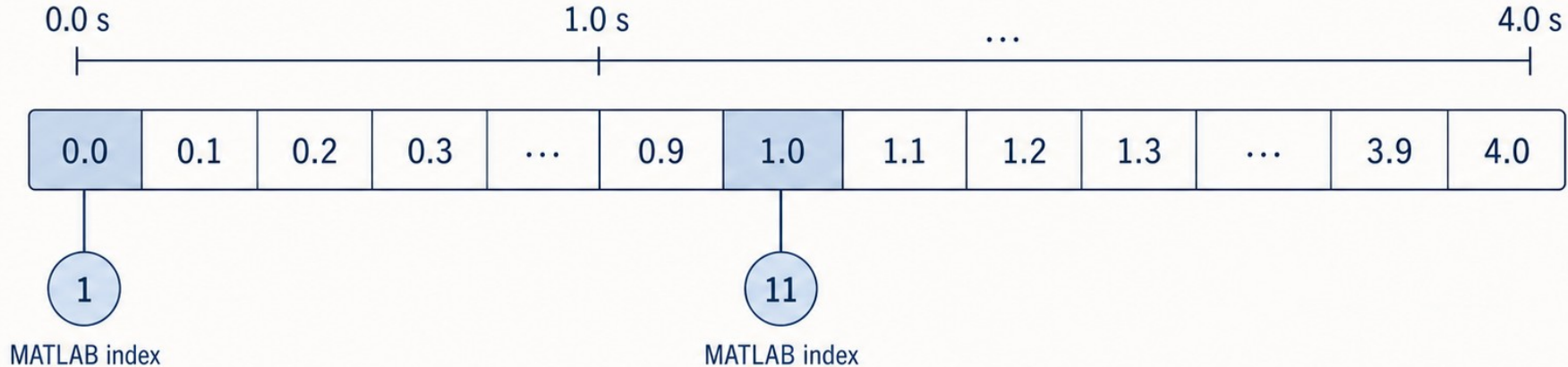


Representation (samples we use):
selected moments from the same model

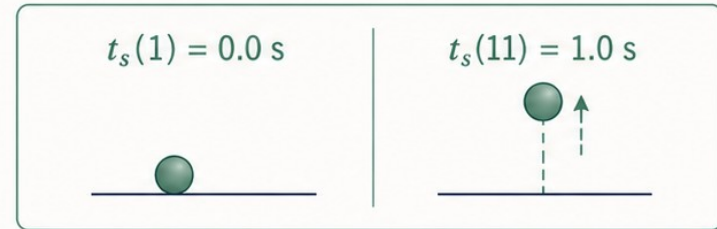
Different sampling → different detail,
same underlying motion.

One index represents one physical moment

Time array: $t_s = 0:0.1:4$ s



- $t_s(1)$ is the launch time.
- $t_s(11)$ selects 1.0 s.
- An index is a storage location, not a time value.



MATLAB indexing starts at 1

Make one equation work for many times

$$y(t) = y_0 + v_0 t - \frac{1}{2} g t^2$$

starting position

$y_0 = 0 \text{ m}$

launch contribution

$v_0 t \rightarrow \text{metres}$

gravity contribution

$-\frac{1}{2} g t^2 \rightarrow \text{metres}$

One stored time $\rightarrow$ one height

t_s [0.0 0.1 0.2 ...]

y_m [0.0 1.95 3.80 ...]

Each stored time gets one height.



Every term on the right has units of metres.

y_0 m

$v_0 t$ m

$\frac{1}{2} g t^2$ m

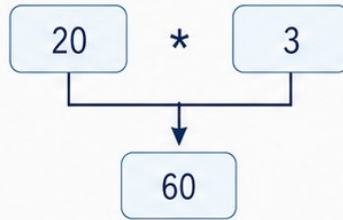
The missing dot changes the operation

The dot means apply the operation element by element.

Scalar multiplication

$20 * 3$

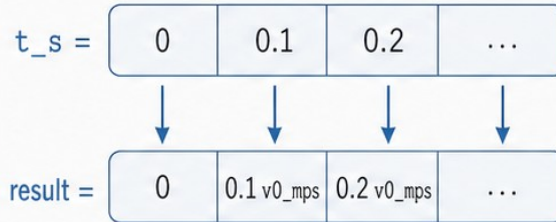
one scalar



$20 * 3$ produces one scalar.

One value per time

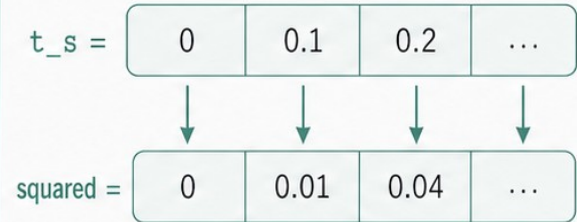
$v0_mps .* t_s$



$v0_mps .* t_s$ produces
one value per time.

Square each stored time

$t_s.^2$



$t_s.^2$ squares every
stored time value.

Checkpoint: Which expression returns an array?

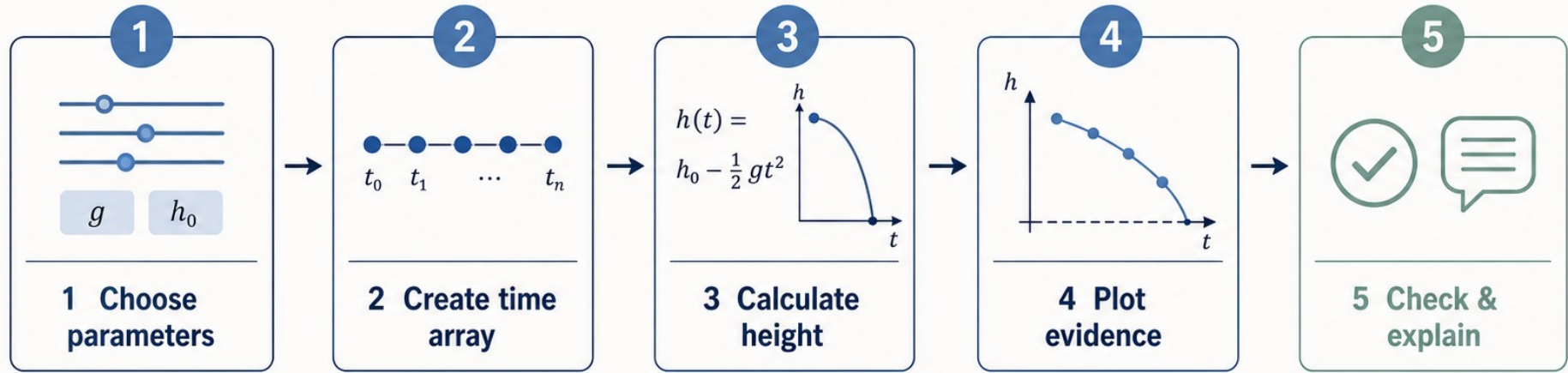


$v0_mps .* t_s$



$t_s.^2$

Build the algorithm before writing syntax



Choose the physical parameters.



Create the time array.



Calculate height at every time.



Plot, check, and explain.

Read the Live Script in the order the physics runs

```
1 — [ v0 = 20;  
      g = 9.81;  
2 — [ t = linspace(0,4,11);  
3 — [ y = v0.*t - 0.5*g.*t.^2;  
4 — [ plot(t,y,'LineWidth',2);  
5 — [ y(1)
```

1 PARAMETERS

Set the launch speed and gravity.

2 ARRAY

Choose the sampled times.

3 CALCULATION

Evaluate height for every time.

4 PLOT

Turn the array into evidence.

5 CHECK

At $t = 0$ s, y must be 0 m.

✓ expected: $y(1) = 0$

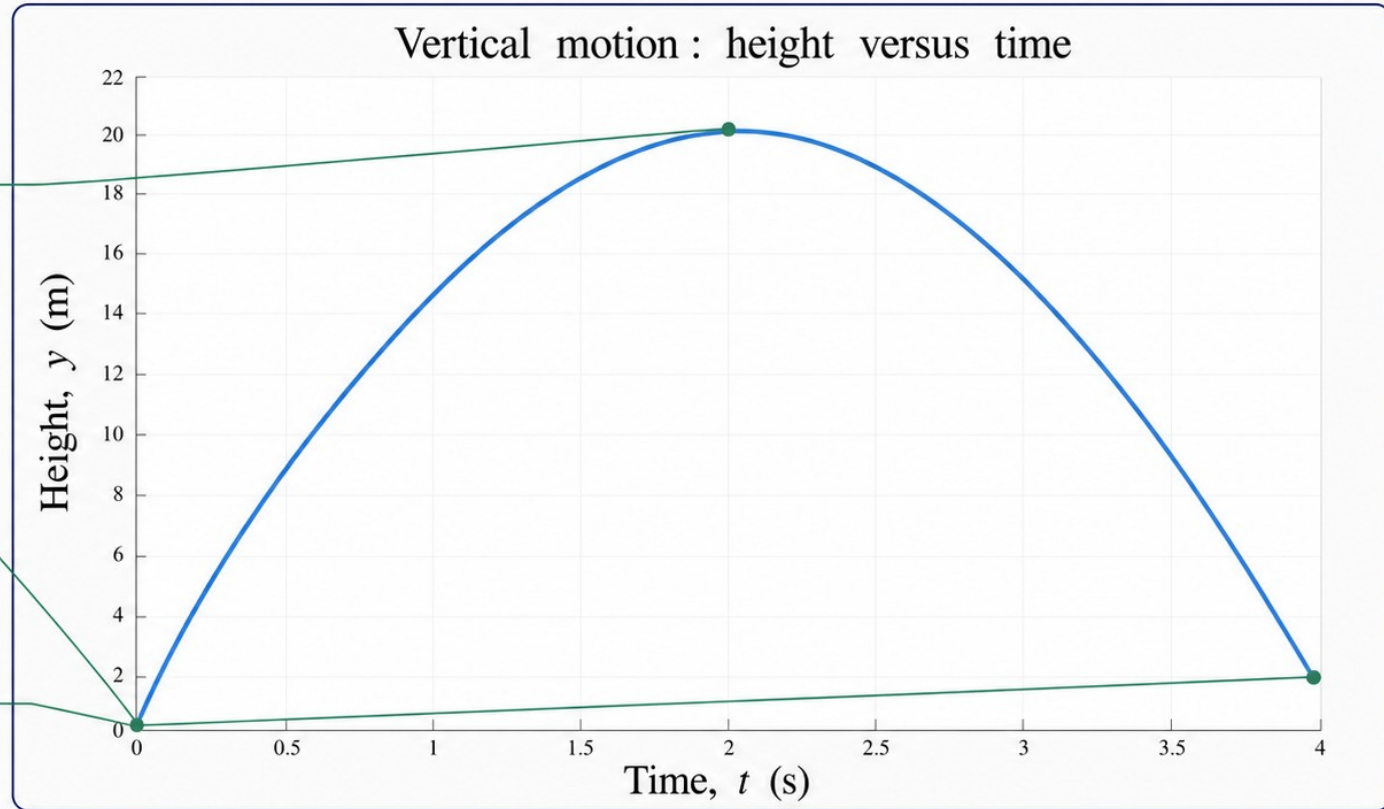
- Read comments and variable names first.
- Trace the array-ready model line.
- Locate the plot and boundary check.

Compare the prediction with the plotted evidence

It rises to
one maximum.

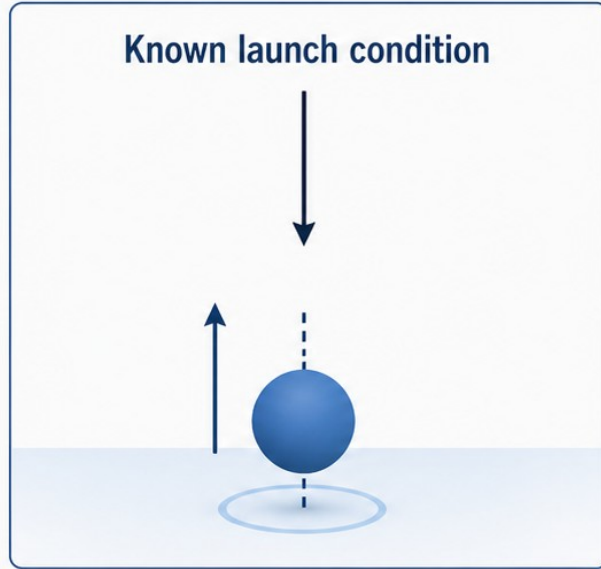
The curve begins
at $y = 0$ m.

It falls toward
the launch height.



Labels and units make the evidence interpretable.

Use a check the model must pass



first stored value

$y_m(1)$

$$t = 0 \text{ s}$$

$$y(1) = 0.000 \text{ m}$$

At $t = 0 \text{ s}$, the model must give $y = 0 \text{ m}$.

Executable boundary check

```
assert(abs(y_m(1)-y0_m) < 1e-12)
```



PASS — launch value matches $y0_m$

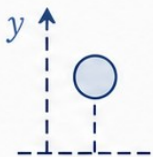
- The known condition is $y(0) = 0 \text{ m}$.
- In a one-based array this is $y_m(1)$.



A smooth graph is not enough: check a known condition.

State what the computation lets us claim

MODEL



$$y(t) = y_0 + v_0 t - \frac{1}{2} g t^2$$

upward positive;
air resistance neglected

SAMPLING

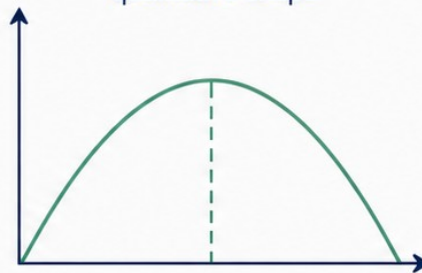


time array: 0:0.1:4 s

sampled positions

GRAPH

qualitative shape



rises to one maximum, then falls

CHECK



$$y_m(1) = 0 \text{ m}$$

starting boundary
condition



Under these assumptions, the ball rises to one maximum and the boundary check supports $y(0) = 0 \text{ m}$.



Not supported: the model proves real motion exactly.

A result needs a unit, a trend, and a limitation.

Week 1 exit ticket

1 Name the output variable and its unit.

2 Explain what $t_s(11)$ selects.

3 Identify the operator that squares every value in t_s .

4 State one graph prediction and one value that can be validated.

Make each computational step traceable.